ELECTROMAGNETIC INDUCTION

TIME: 60min **MARKS: 40**

SECTION I

[*All* *questions are compulsory*]

Question no.1: **[5]**

[i] Lenz law is considered to be ..... law of Faraday.

[a] same as

[b] opposite to the

[c] directional

[d] none of the above.

[ii] The energy conversion taking place in DC motor is.....

[a] Kinetic energy to electrical energy

[b] electrical energy to mechanical energy

[c]mechanical energy to electrical energy

[d] electrical energy to mechanical kinetic rotational energy.

[iii] In a step-up transformer the number of turns in secondary coil are....

[a] equal to number of turns in primary coil

[b] less than number of turns in primary coil

[c] more than number of turns in primary coil

[d] square root of number of turns in primary coil.

[iv] Losses in transformer include....

[a] Hysteresis loss

[b] copper losses

[c] eddy current losses

[d] all the above.

[v] AC dynamo has concave shaped pole surfaces to produce.....

[a] radial magnetic field

[b] parallel magnetic field

[c] cylindrical magnetic field

[d] spherical magnetic field.

Question 2:

Fill in the blanks and rewrite the complete sentence by underlining the blanks: [5]

[i] Transformer works on the principle of ..... [self-induction / mutual induction]

[ii] AC generator has ..... [slip / split] rings.

[iii] Transformer ratio for step down transformer is .... [less than 1 / more than 1]

[iv] DC motor uses ..... [Flemming left hand rule / Flemming right hand rule] to give the direction of force experienced by a conductor.

[v] Electric bell works on the principle of ..... [electromagnetic induction / self- induction].

Question 3:

[a] State the Faraday's law of electromagnetic induction? **[2]**

[b] If a soft iron piece is placed inside a current carrying coil, what is the name given to such a device. Give any one use of the same. **[2]**

[c] Why does the magnetic needle show a deflection of the coil when brought close to a current carrying conductor. If a wire is bent into a circle and carries current in anticlockwise direction, which polarity does this face exhibit? **[2]**

[d] Define the term transformation ratio or turns ratio. **[2]**

[e] State the function of commutator in case of a DC motor. **[2]**

SECTION II

[*Attempt* ***any two*** *questions out of three***]**

Question 4:

[a] Draw a neat labeled diagram of magnetic lines of force in a straight current carrying conductor. **[3]**

[b] Name three factors on which the magnitude of force on a current carrying conductor placed in a magnetic field depends. **[3]**

[c] Show by means of a diagram the effect of motion of magnet in a closed circular coil when either the north pole enters or leaves the magnet or the south pole enters or leaves the coil. Name the rule used to justify your observations. **[4]**

Question 5:

[a] Give three uses of an electromagnet that makes it more preferred over the use of permanent magnet. **[3]**

[b] Draw a neat labeled diagram of a DC motor and state the principle on which it works. **[4]**

[c] State the function of the following in case of DC motor:

[i] Split rings [ii] Concave shaped poles [iii] Armature. **[3]**

Question 6:

[a] A magnetic needle is placed below a current carrying conductor PQ with its end Q pointing towards the north and end P pointing towards the south pole of the earth's magnetic field. The conductor PQ is also connected to a cell with end Q connected to the positive terminal and key. Answer the following questions:

[i] in which direction will the needle deflect if the key is closed?

[ii] Why is this deflection produced?

[iii] What would be the change in deflection if any if the needle is kept above the conductor carrying current instead of below?

[iv] Name one device which works on this principle. **[4]**

[b] State three methods of minimizing losses in a transformer **[3]**

[c] Draw the magnetic lines of force in a solenoid carrying current. Also mark the polarities at both ends of the solenoid. **[3]**

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